



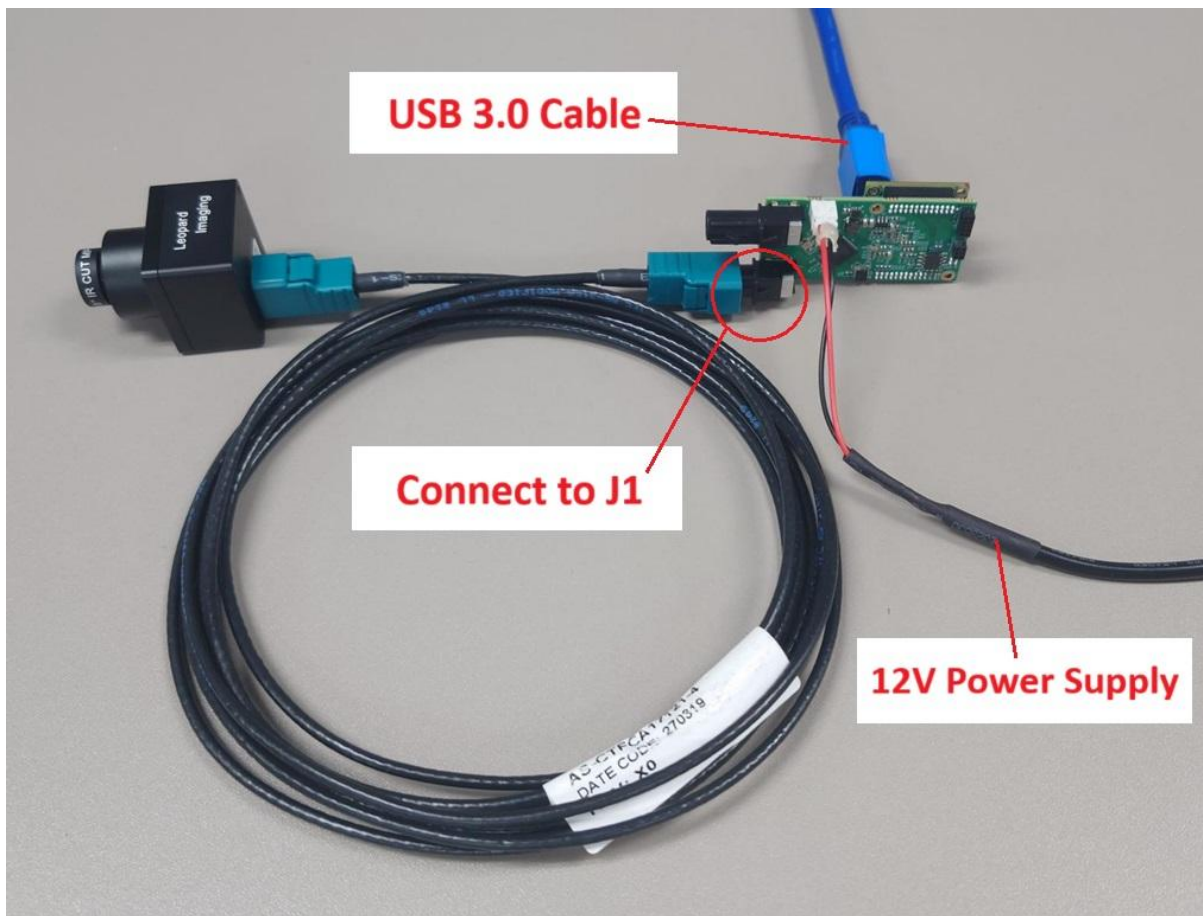
LI-USB30-AR0820-GMSL3_User_Guide_20250618

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Overview

This firmware is for LI-USB30-AR0820-GMSL3 camera.
This firmware supports 3840 x 2160
This camera doesn't include ISP and outputs RAW data.

Platform	Camera
1 x LI-GMSL3-USB	1 x LI-AR0820-GMSL3
Cable	Adapter/Carrier Board
1 x USB 3.0 Micro-B cable 1 x FAK-SMZSMZ Fakra cable	1 x 12VDC power supply





LI-USB30-AR0820-GMSL3_User_Guide_20250618

Revision	SVN version	Release Date	Author	Tested By
2025_06_18		05/28/2024	Shelby Hache	Shelby Hache
Updates				
Revision	Description			Release Date
2020_05_28	First Release			05/28/2020
2025_06_18	Updated Instructions and Images			06/18/2025
Known bugs				



Setup Procedure 1/3

- Hardware:

1. LI-AR0820-GMSL3 x 1
2. LI-GMSL3-USB Tester x 1
3. FAK-SMZSMZ Fakra cable x 1
4. USB 3.0 Micro-B cable x 1
5. LI-PS12-01 x 1

- Hardware Setup:

Connect the camera and USB 3.0 Tester board as the picture on the first page.

- Software:

This camera kit can be tested with any one of the below software:

1. Camera tool (Windows OS)

The Camera tool can be downloaded from the link below:

https://www.dropbox.com/s/8daqfypyhnbj5/CameraUSB30_3_2_20190622.7z?dl=0

It's better to use the 7-zip to uncompress the package:

<http://www.7-zip.org/download.html>

And install the software below on your PC:

https://www.dropbox.com/s/6uswl40z8rqh2et/vcredist_x86.exe?dl=0

If needed, you can also download the camera tool SDK:

https://www.dropbox.com/s/j6ccl8cvt75gu2g/USB30_CameraTool_SDK_rev1440_20190622.7z?dl=0

2. Linux Camera tool (Linux OS)

The Linux camera tool can be downloaded from link below.

https://www.dropbox.com/s/4m2efo696px9739/linux_camera_tool-master_20190624.zip?dl=0

For how to install it, please refer to the README.md in below link. (the latest version of Linux camera tool may have an issue, so please use above Linux camera tool)

https://github.com/LI01/linux_camera_tool

Please follow the README.md in GitHub to install the Linux camera tool.

There are instructions online for how to install the OpenCV on Ubuntu OS.

Below is an example:

<https://www.learnopencv.com/install-opencv-3-4-4-on-ubuntu-18-04/>



Run Camera 1/3

- Camera Tool (Windows OS)

1. Connect the camera to port J1 of USB 3.0 Tester. (refer to picture in page 1)
2. Plug in the 12V power supply to USB 3.0 Tester.
3. Connect the camera to your PC (**USB3.0 port**) using the USB 3.0 cable.
4. Open camera Tool.

Please make sure to complete these steps in numerical order. Steps 2 and 3 cannot be reversed.

Color matrix:

UPDATE CLOSE

Camera

DefaultMode: 0 ☐ FocusProcDownSample

RAW Interpolation

☐ GammaEna Gamma: 1.0

☒ RGBGainOffsetEna

Red Gain: 512 Red Offset: 0

Green Gain: 250 Green Offset: 0

Blue Gain: 512 Blue Offset: 0

☐ RGB2RGBMatrixEna

Red Green Blue

Red 256 0 0

You can adjust the color matrix in
Options→ Configuration.

Register access function:

CameraTool

File Devices DevInfo

Device: LI-AR0820 HW_Rev

I2C Access

I2C Addr 0x20

I2C Addr Width

☐ 8 bits ☒ 16 bits

I2C Data Width

☐ 8 ☒ 16 ☐ 32

Reg Addr 0x3012

Reg Value 0x1100

Read Write Hex

Batch Cmd

Load Exec Clear

This camera supports register access function. The I2C address is **0x20**

To select the exposure setting, use 0x3012 for the Register Address, and the Register Value can be changed to adjust the exposure.

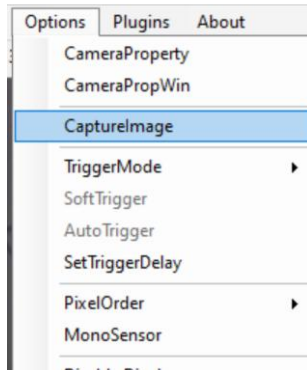
You can also use BatchCmd.txt file to write/read register.



Run Camera 2/3

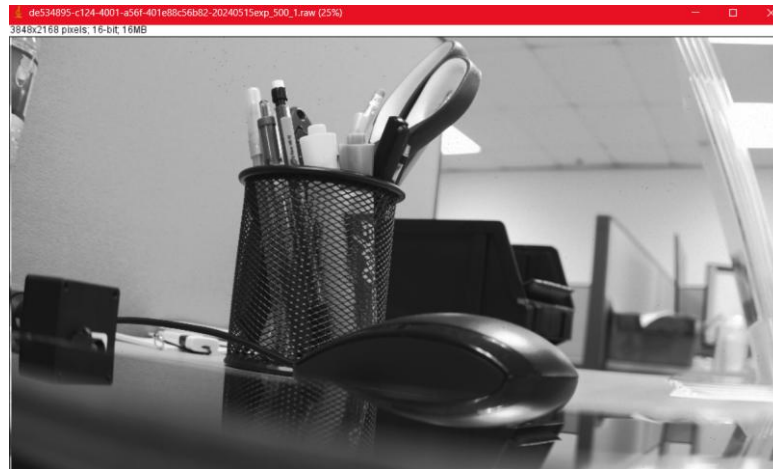
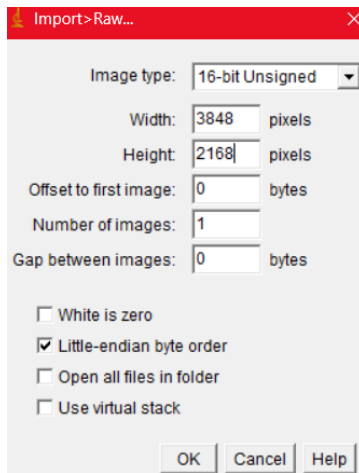
Capture Image:

The images can be captured by clicking Options→ CaptureImage. Two images (RAW and BMP) will be saved to PC.



The raw image can be viewed by ImageJ.

<https://www.dropbox.com/s/fsvfndy6s9ft03i/ImageJ.7z?dl=0>



Note: This camera tool supports Raw-to-RGB function which will reduce the display frame rate. You can click Options→ DisableDisplay to get actual frame rate from the sensor. You can also use other regular software (like AMcap) to get higher frame rate, but the video will be green (like below) due to lack of Raw-to-RGB conversion function.



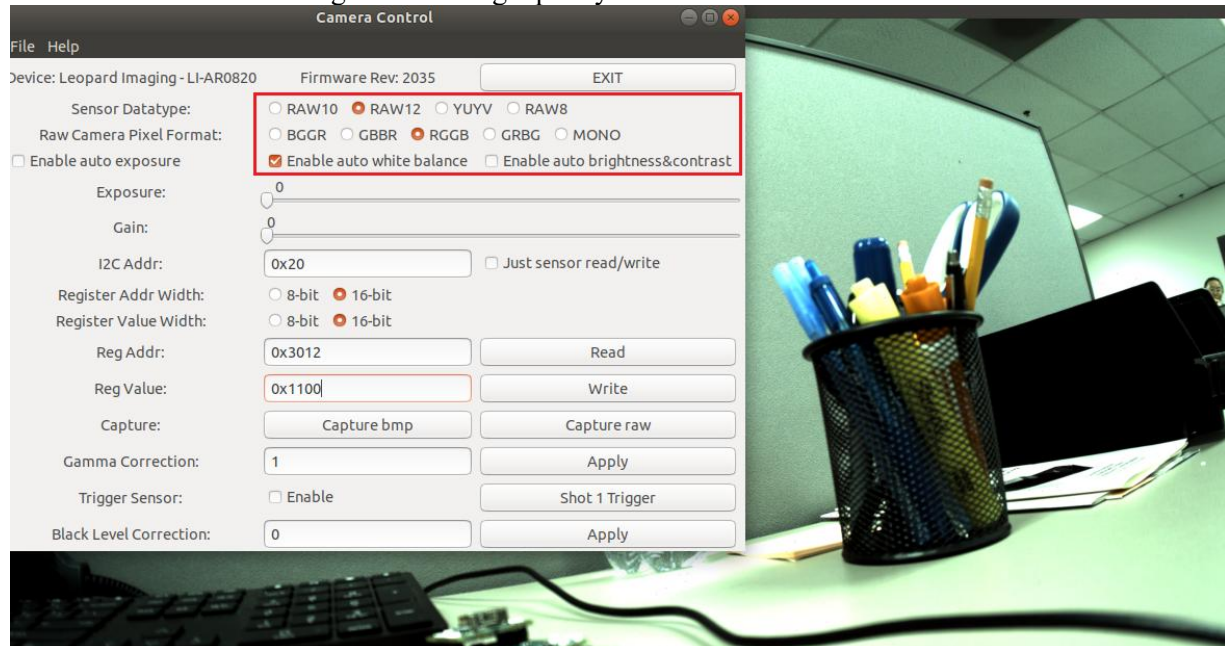
Run Camera 3/3

- Camera Tool (Linux OS)

1. Connect the camera to J1 of USB 3.0 Tester. (refer to picture in page 1)
2. Plug in the 12V power supply to USB 3.0 Tester.
3. Connect the camera to PC (**USB3.0 port**) through USB 3.0 cable.
4. Open a terminal and use the command “leopard_cam” to open the camera.

Please make sure to complete these steps in numerical order. Steps 2 and 3 cannot be reversed.

Set the camera as below to get better image quality.



This camera tool can be used to write/read registers and capture Raw and BMP images.

I2C Addr:	0x20	☐ Just sensor read/write
Register Addr Width:	☐ 8-bit ☒ 16-bit	
Register Value Width:	☐ 8-bit ☒ 16-bit	
Reg Addr:	0x3012	Read
Reg Value:	0x1100	Write



Note 1/3

This camera kit is pre-loaded with Firmware (in USB 3.0 Tester) and FPGA (in USB 3.0 Tester). If there are any new firmware and/or FPGA binary from Leopard Imaging, you can refer to below instructions to update them.

1. Firmware Update:

Please use the **LP_USB3_FirmwareUpdateTool** in the camera tool folder to update the firmware.

- 1) Click “Erase” to erase the old firmware.
- 2) Click “FW Update” button to select the lif file. (If the “FW_Update is unavailable, please install the WestBridge driver, check below)
- 3) The update process may take about 15 seconds.
- 4) If the process takes too long, please disconnect the USB and reconnect it to PC.
Then try the update tool again.

——Install WestBridge:

If the camera cannot be recognized after you update the firmware, and there is a device name “WestBridge” on the Device Manager, please download the driver from the link below and install it.

https://www.dropbox.com/s/4yx2p31b7qo2gix/WestBridge_driver.zip?dl=0

- 1) Right click on “WestBridge” and select Update Driver Software.
- 2) Choose browse my computer for driver software.
- 3) Click Browse, locate the driver at the downloaded and unzipped folder. (C:\temp\driver\bin\ for example)
If your PC has Win7 or later version, please select the folder “win7”.
- 4) Click next and complete the installation process.

After install the driver, please update the firmware again.



Note 2/3

2. FPGA Update:

FPGA Binary: mipi2cmos_4lane_raw12_v1.3.3_rstd15.rbt

1) Download the Lattice programmer tool software **Programmer Standalone 3.10 64-bit for Windows** from link below and install it to your PC.

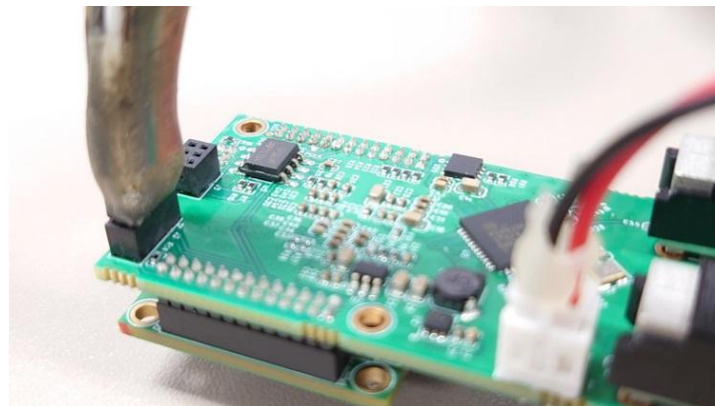
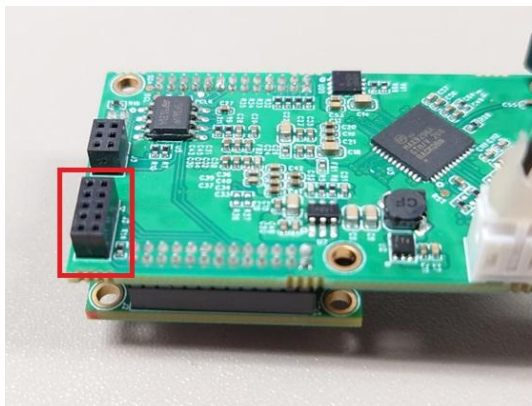
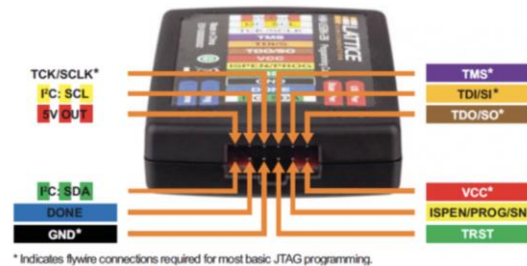
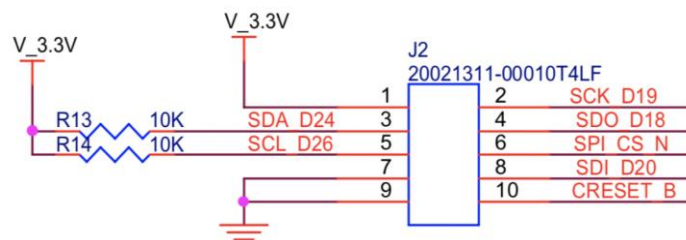
<http://www.latticesemi.com/programmer>

2) Connect the HW-USBN-2A or HW-USBN-2B to J2 of LI-MAX9296-DESER GMSL3 board. Below is the pinout. You may need to build an adapter cable for the programmer tool and J2. J2 is a 1.27mm pitch interface.

Below pins in red need to be connected.

Only below pins in red need to be connected:

TI954 Board <— —> HW-USBN-2B	TI954 Board <— —> HW-USBN-2B
Pin1: V_3.3V <— —> VCC	Pin2: SCK_D19 <— —> TCK/SCLK
Pin3: N/A	Pin4: SDO_D18 <— —> TDO/SO
Pin5: N/A	Pin6: SPI_CS_N <— —> ISPEN/PROG
Pin7: GND <— —> GND	Pin8: SDI_D20 <— —> TDI/SI
Pin9: N/A	Pin10: CRESET_B <— —> TRST

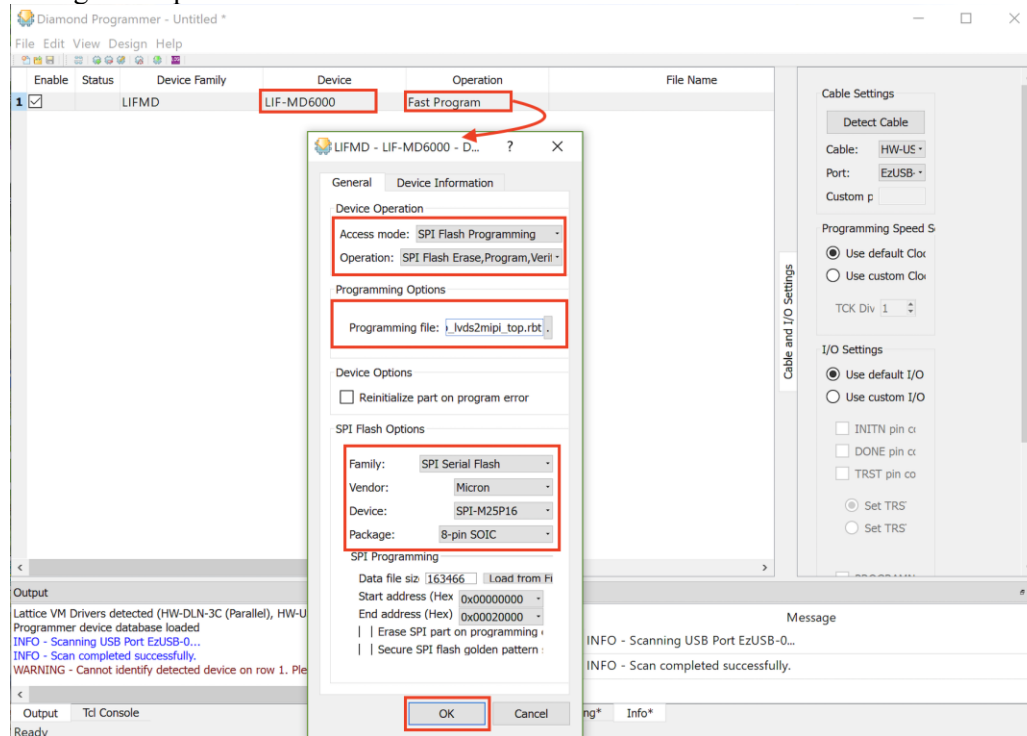




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Note 3/3

- 3) Connect the USB3.0 Tester to your PC via USB 3.0 cable.
- 4) Open **LP_USB3_FirmwareUpdateTool** in the camera tool folder and click “Erase” to erase the firmware.
- 5) Open Lattice software (which you installed in 1). The Device name should be “LIF-MD6000”. Set the settings of “Operation” as below and click “OK”.



- 6) Click program icon.



- 7) It may take 9s to program the FPGA.



- 8) Refer to the “Firmware Update” section to install the firmware.